Experiment 4: guide for the lab

10060 Physics (Polytechnical Foundation) · Friday 23 October 2026

Equipment from 13:00 · report due 17:00 · files: this PDF, `Experiment4_analysis.ipynb`, `Experiment4_Dummy_Report.pdf`

1 The short version

The body. The iOLab with a tape roll pressed onto each end. It rolls on the two tape rolls, so it turns about the y axis of the iOLab, and the gyroscope measures that angular velocity ω directly. You measure the moment of inertia of the whole body, then subtract the two tape rolls.
Method N, rotational Newton's second law. Push the body up the ramp and let it roll back. The gyroscope shows $\omega(t)$ as two straight lines. Their slopes are α_{up} and α_{down}. Rolling resistance makes α_{up} a little larger than α_{down}, and their mean has no friction in it. With torques about the contact line P:

$$I_P = \frac{MgR\sin\theta}{\bar{\alpha}}, \qquad \bar{\alpha} = \tfrac{1}{2}(\alpha_{\text{up}} + \alpha_{\text{down}}).$$

Method E, energy. Release the body from rest at a gate and let it roll the distance s_{AB} to the end of the board. Read ω_B there. Energy conservation with the friction loss ε from Method N:

$$I_P = \frac{2Mg\,s_{AB}\sin\theta\,(1-\varepsilon)}{\omega_B^2}.$$

Both. The parallel axis theorem gives the body, $I = I_P - MR^2$, and $I_{\text{iOLab}} = I - I_{\text{rolls}}$ with $\frac{1}{2}m(r^2 + R^2)$ for each tape roll.
The catch. The iOLab is only about 5 % of I_P. A 1 % error in I_P becomes an 18 % error in I_{iOLab}. Three measurements decide the result, and most of this guide is about them: the angle (accelerometer flipped on the board, against an upright reference), the gyroscope scale (roll the body three full turns along the table) and friction (up and down runs). Expect I_{iOLab} between 1×10^{-4} and 2×10^{-4} kg m^2.

What you hand in at 17:00

- One lab report of at most 3 pages, then page 4 with the reflections (3 to 4 lines for each of the three questions).
- A hand drawn sketch: the ramp seen from the side, and the body seen from the front. Draw on paper, photograph it from straight above, crop it.
- Every measured value with its uncertainty and its unit.
- The DTU Learn upload folder is set up per group. Agree now who uploads.
- AI tools are allowed for the report this semester. Every number in it must still come from your own data, and you must be able to explain every formula.

What to bring

Laptop with the iOLab app and Python (numpy, pandas, matplotlib), the notebook with its `example_data` folder, this guide. A 2 m tape measure and a 30 cm ruler, a caliper if you can borrow one, a strip of paper about 2 cm by 40 cm, masking tape, a pen, a towel or a folded jumper as a soft catch, a set square (a book with a square corner also works), paper for the sketch and the run log, and a phone for photos. Charge the laptop.

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2 Test run before Friday

You have an iOLab and two tape rolls at home, so you can test the whole chain before the lab. It takes about 30 minutes. Send the files to me; I will run them through the notebook and fix anything that does not read well before Friday.

1. **Gyroscope check.** Tick only Gyroscope. Write down the rate in the chart title, for example "Gyroscope (380 Hz)". Check that the unit is rad/s. Spin the iOLab fast by hand, or flat on the table like a coin. If ω goes above 9 rad/s without a flat top, you have the large range ($\pm$34.9 rad/s) and all is well. A flat top lower down is the range limit (Section 6).
2. **Angle recording**, rolls still off. Tick only Accelerometer. Hold the bare iOLab upright against a wall, y up, 4 s; then upside down, 4 s. Then lay it on its back on a board propped on books at 10° to 15°, y up the slope, 4 s; turn it around, y down the slope, 4 s.
3. **Calibration roll.** Press the rolls on, not past the notches. Draw a pen line across the edge of one roll and put a mark on the table where it touches. Tick Gyroscope only. Record, 1 s still, then roll the body slowly by hand in a straight line for exactly 3 full turns, until the pen line touches the table for the third time. Mark that point, 1 s still, Stop. Measure the distance between the marks.
4. **One Method N run** on the same board. Body at rest near the bottom, 1 s still, one short push up the board, let it roll up and back, catch it before the end.
5. **One Method E run.** Body at rest against a ruler held across the board, 1 s still, lift the ruler, let the body roll off the lower end onto a towel.

Export the accelerometer chart for recording 2 and the gyroscope chart for recordings 3 to 5 (toolbar icon at the bottom left of the chart, then Export to CSV; the files land in `Documents/iOLab-WorkFiles/export`). Send the five files with a note on which is which, the 3 turn distance, the board length and the two heights, and one photo of the body with the rolls on.

The test answers six questions you cannot afford to find out on Friday: the gyroscope rate, its unit, its range, the export format, whether ω_y really is the large axis, and how big the scale error k of your gyroscope is.

3 The rubric, line by line

The Level 4 column is cut off in the rubric PDF. The wording below follows Level 3 without its faults, the same pattern as the Experiment 3 rubric, where Level 4 was printed.

Rubric line	Level 4 (expected wording)	What you do
Sketch of the setup	A very good and clear sketch.	Two hand drawn views: the ramp from the side (θ, gate, A, B, s_{AB}, towel, books, laptop) and the body from the front (iOLab, both rolls, r, R, the y axis). Section 9.3 lists every label.
Collect and represent data	All data collected, organized and presented clearly.	One table of the measured inputs, one table per method with every run and the mean, one figure with a run of each method.
Uncertainties on measured quantities	All uncertainties present and correctly estimated, with units.	For every input say where its uncertainty comes from: scale step, $\sigma/\sqrt{N}$, SDOM, propagation. The budget shows which input dominates.
Data analysis to determine a quantity	The moment of inertia determined correctly in several ways.	Two methods, each giving I_{iOLab} with an uncertainty, a standard score between them, and a comparison with the box model. Show the formulas.
Purpose and communication	Clear purpose, good communication, all measurements with units.	Purpose in two sentences. The windmill problem in one sentence. The same symbols everywhere.
Reflections	(not described)	Page 4. Three questions, concrete details from the day.
Overall judgement	(not described)	Correct physics, tidy layout, nothing padded, inside the page limit.

4 Time plan, 13:00 to 17:00

Time	Block	Tasks
13:00 to 13:15	Start and calibrate	Connect the iOLab, Reset twice, run the Acc/Magn/Gyro calibration. Gyroscope check (rate, unit, range, axis). One person starts the sketch.
13:15 to 13:30	The parts	Rolls still off: weigh the iOLab and each roll, measure the radii and the iOLab size.
13:30 to 13:50	Ramp and angle	Build the ramp. One accelerometer recording with the bare iOLab: upright reference, then three places along the board. Table check. Ruler heights. Place the gate mark and measure D_{gate}.
13:50 to 14:10	Rolls on, calibration	Mount the rolls, weigh the whole body, check that it rolls straight. Three calibration rolls of three turns each. Export after every roll.
14:10 to 14:35	Method N	6 to 8 up and down runs. Export after every run.
14:35 to 14:55	Method E	6 to 8 releases from the gate. Export after every run.
14:55 to 15:15	Analysis	Run the notebook. Look at every plot. Redo bad runs now, while the ramp still stands. Do not move the ramp before this check.
15:15 to 16:25	Pages 1 to 3	Split: one person on setup and method, one on tables and figures, one on theory and discussion. Swap and read each other's parts at 16:05.
16:25 to 16:40	Page 4	Reflections.
16:40 to 16:55	Final check	Checklist in Section 9.8. Export to PDF.
16:55	Submit	Upload on DTU Learn. Do not wait until 17:00.

If you run late, cut runs, not checks. Five runs per method with the calibration roll and the angle recording are worth more than ten runs without them.

5 Physics and formulas

5.1 The body and what the gyroscope measures

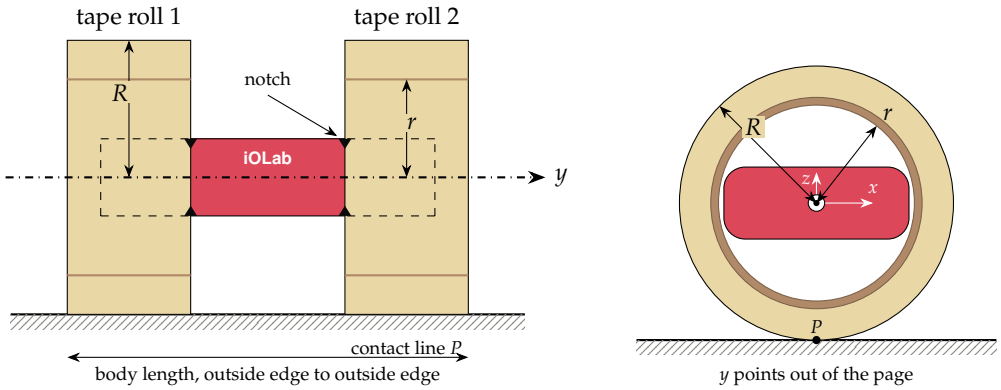


Figure 1. The body. Left: seen from the front, the two tape rolls sit on the ends of the iOLab up to the notches (dashed: the parts of the iOLab hidden inside the cardboard cores). Right: seen along the axis. The iOLab end sits inside the core of inner radius r; the body rolls on the outer surface, radius R. The rolling axis is the y axis of the iOLab.

The iOLab is about 130 mm long (y), 75 mm wide (x) and 30 mm thick (z). Its ends fit inside the cardboard cores of the tape rolls. When the body rolls, it turns about the common axis of the two rolls, which is the y axis of the iOLab when the rolls sit straight. The gyroscope measures the angular velocity about all three axes; you only need ω_y, and ω_x, ω_z stay near zero. The wheel sensor is useless today, because the wheels of the iOLab touch nothing.

Symbol	Meaning
M	total mass of the body: iOLab plus both rolls
R	rolling radius: outer radius of the tape rolls
m_j, r_j, R_j	mass, inner radius (inside of the core) and outer radius of roll j
I	moment of inertia of the whole body about the rolling axis
$I_P = I + MR^2$	moment of inertia about the contact line P, where the rolls touch the board
$I_{\text{rolls}} = I_1 + I_2$	the two tape rolls
$I_{\text{iOLab}} = I - I_{\text{rolls}}$	the iOLab: the quantity you must find
θ	ramp angle against the horizontal
ω, α	angular velocity and angular acceleration, counted positive when the body rolls down the ramp

5.2 Rolling without slipping

When the rolls do not slip, the contact line P is at rest for an instant. The centre then moves with

$$v = \omega R, \qquad a = \alpha R. \tag{1}$$

The kinetic energy is the energy of the centre plus the energy of turning about the centre:

$$K = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I\omega^2 = \tfrac{1}{2}\left(I + MR^2\right)\omega^2 = \tfrac{1}{2}\,I_P\,\omega^2. \tag{2}$$

$I_P = I + MR^2$ is the parallel axis theorem (University Physics Vol. 1, Section 10.5): the moment of inertia about a line parallel to the axis through the centre of mass, a distance R away. At each instant, rolling is pure rotation about P.

5.3 Method N: torque about the contact line

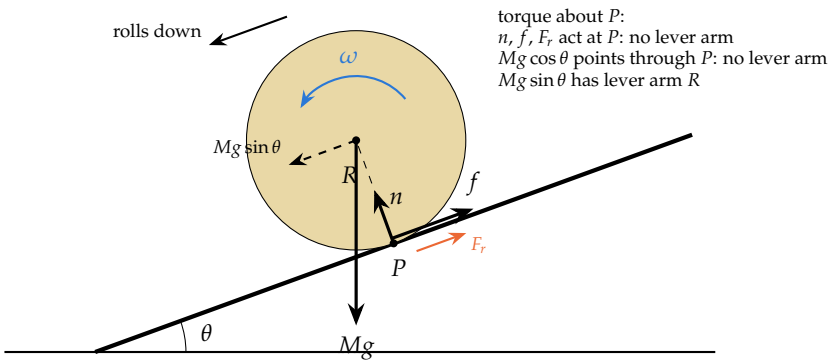


Figure 2. Forces on the body while it rolls down. f is static friction, which makes it roll. F_r is rolling resistance: it always points against the motion, so it flips direction at the turning point. Taking torques about the contact line P leaves only $Mg\sin\theta$ with lever arm R.

The board pushes on the rolls at P (normal force n and friction), and gravity pulls at the centre. Take torques about P. Every force that acts at P has no lever arm. Split gravity: $Mg\cos\theta$ points straight through P, and $Mg\sin\theta$ points along the board with lever arm R. The rotational form of Newton's second law (University Physics Vol. 1, Section 10.7) about P:

$$\sum \tau_P = MgR\sin\theta = I_P\,\alpha \qquad \Longrightarrow \qquad \alpha = \frac{MgR\sin\theta}{I + MR^2}. \tag{3}$$

The same result comes from the centre of mass: $Ma = Mg\sin\theta - f$ and $I\alpha = fR$ with $a = \alpha R$. Eliminate f and you get (3). Use whichever derivation you like in the report; the one about P is the shortest.

Friction cancels in up and down runs. Rolling resistance acts like a small force F_r at P that always opposes the motion. Rolling down, it points up the ramp; rolling up, it points down the ramp, the same way as $Mg\sin\theta$. With α counted positive in the rolling down sense (going up the body slows down, coming down it speeds

up, so both are positive):

$$\text{up:} \quad I_P\,\alpha_{\text{up}} = MgR\sin\theta + F_r R, \tag{4}$$

$$\text{down:} \quad I_P\,\alpha_{\text{down}} = MgR\sin\theta - F_r R. \tag{5}$$

Add the two:

$$\boxed{I_P = \frac{MgR\sin\theta}{\bar{\alpha}}, \qquad \bar{\alpha} = \tfrac{1}{2}\left(\alpha_{\text{up}} + \alpha_{\text{down}}\right)} \tag{6}$$

Subtract them to get the size of the friction, as a fraction of $Mg\sin\theta$:

$$\varepsilon = \frac{F_r}{Mg\sin\theta} = \frac{\alpha_{\text{up}} - \alpha_{\text{down}}}{\alpha_{\text{up}} + \alpha_{\text{down}}}. \tag{7}$$

Then $I = I_P - MR^2$ and $I_{\text{iOLab}} = I - I_{\text{rolls}}$. This is the trick from Experiment 3 Part 2, now with the gyroscope instead of the wheel. Method E needs ε.

5.4 Method E: energy from the gate to the end of the board

The body starts at rest with its contact line at A and rolls the distance s_{AB} along the board to the end B. Its centre drops $\Delta h = s_{AB}\sin\theta$. While it rolls without slipping, static friction does no work (it acts at a point at rest). Rolling resistance takes $W = F_r\,s_{AB} = \varepsilon\,Mg\,\Delta h$. With (2):

$$Mg\,\Delta h = \tfrac{1}{2} I_P\,\omega_B^2 + \varepsilon\,Mg\,\Delta h \qquad \Longrightarrow \qquad \boxed{I_P = \frac{2Mg\,s_{AB}\sin\theta\,(1-\varepsilon)}{\omega_B^2}} \tag{8}$$

Where the contact line is. The gate is a ruler held on its edge across the board, square to the board surface. It touches the roll at the height of the axis, a distance R further down the board than the contact line. So

$$s_{AB} = D_{\text{gate}} + R, \tag{9}$$

where D_{gate} is measured along the board from the gate face to the end of the board. Forgetting the $+R$ is a 10 % error in s_{AB}.

Why ω_B is easy to read. At B the body leaves the board. In the short flight to the table no torque acts about the centre, so ω stays constant for about 0.04 s. The graph of ω shows a sharp corner: a straight rise, a short flat part, then a small drop when the body lands (Figure 8). ω_B is the value at the corner.

Air drag, a small correction. Drag grows with v^2. At the end of a 0.5 m run it is about 1 % of $Mg\sin\theta$. The energy runs end faster than the average speed in the Method N windows, so they lose a bit more to drag than ε contains. With $F_d = k_d v^2$ and $k_d = \frac{1}{2}\rho C_D A$ ($\rho = 1.2\,\text{kg/m}^3$, $C_D \approx 1$, $A = 2R$ times the body length) the notebook replaces $(1-\varepsilon)$ by $(1-\varepsilon-\Delta f)$, with

$$\Delta f = \frac{k_d\,v_B^2}{2Mg\sin\theta} - \frac{k_d\langle v^2\rangle_N}{Mg\sin\theta}, \tag{10}$$

the drag work in the energy run minus the drag share already inside ε. C_D is only an estimate, so half of Δf is its uncertainty. In the example data $\Delta f = 0.003 \pm 0.002$. One sentence in the report is enough.

Two free checks. The slope of the straight rise in an energy run is α_{down}; it should match Method N (slightly smaller, because of the extra drag). And R times the gyroscope angle from release to the corner must equal s_{AB}. If it does, the body did not slip and your distances are right.

5.5 The tape rolls and the iOLab

A tape roll is a thick hollow cylinder (University Physics Vol. 1, Figure 10.20):

$$I_j = \tfrac{1}{2}\,m_j\left(r_j^2 + R_j^2\right), \qquad I_{\text{rolls}} = I_1 + I_2. \tag{11}$$

Measure each roll by itself. The formula treats the cardboard core as if it were tape. Cardboard is lighter, so the real mass sits a little further out and the real I_j is about 1 % larger. In the example data that shifts I_{rolls} by $0.07 \times 10^{-4}\,\text{kg m}^2$, which would make I_{iOLab} that much too large. The notebook puts this shift into the uncertainty budget as "roll model".

For the iOLab itself a box model tells you what to expect. Seen along y, the iOLab is a rectangle $w \times t$ (75 mm by 30 mm). Two limits:

$$\text{mass spread evenly: } I = \tfrac{1}{12} m \left(w^2 + t^2\right), \qquad \text{all mass in the walls: } I = \tfrac{1}{12} m\,(w+t)^2. \tag{12}$$

The first is the slab in Figure 10.20. The second follows from adding four thin walls, each with the parallel axis theorem. With $m = 0.196\,\text{kg}$ they give 1.07×10^{-4} and $1.81 \times 10^{-4}\,\text{kg m}^2$. The real iOLab has its batteries and circuit board near the middle and a plastic shell outside, so it lies between the two. This is your expectation before the lab, and reflection question 1 asks for it.

5.6 Why the iOLab is hard to measure

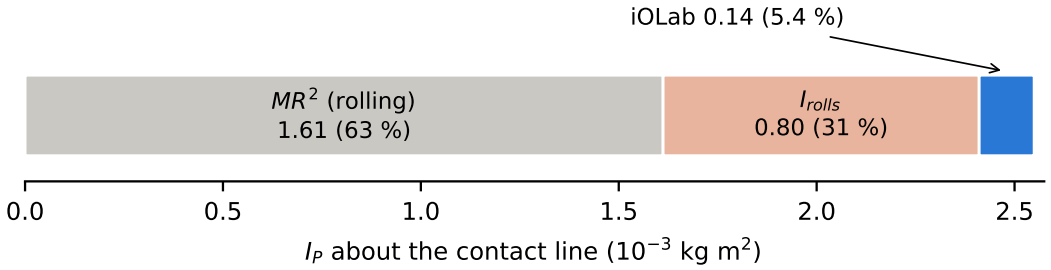


Figure 3. What I_P is made of, for the example data. You measure the whole bar; the iOLab is the small blue end.

Both methods measure I_P. It is made of MR^2 (63 %), the tape rolls (31 %) and the iOLab (5 %). $I_{\text{iOLab}} = I_P - MR^2 - I_{\text{rolls}}$ is a small difference of large numbers, so every relative error in I_P is multiplied by $I_P / I_{\text{iOLab}} \approx 18$:

A 1 % error in	Method N	Method E
$\sin\theta$	18 % in I_{iOLab}	18 % in I_{iOLab}
gyroscope scale k	18 %	36 % (ω^2)
$\bar{\alpha}$ or ω_B	18 %	36 %
R	5 %	21 %
M	7 %	7 %
I_{rolls}	6 %	6 %
friction ignored ($\varepsilon \approx 0.017$)	cancels	31 %

So the targets are: $\sin\theta$ to 0.15 %, k to 0.1 %, R to 0.1 %, friction cancelled. All four are reachable with the steps in Section 7. This table is also the core of your discussion: it explains why a careful measurement still ends with an uncertainty of 5 to 10 %.

5.7 The gyroscope scale and the rolling radius: the calibration roll

A gyroscope can be off by 1 to 3 % in its scale. The calibration in the Tools menu only sets its zero. So you calibrate it against a rotation you know: roll the body by hand along the table through exactly $N = 3$ turns, pen line to pen line, with 1 s still before and after.

$$k = \frac{\Phi}{2\pi N}, \qquad \Phi = \int \omega_y\, dt \text{ (offset removed)}; \qquad R = \frac{d}{2\pi N}, \tag{13}$$

where d is the distance between the two table marks. Divide every gyroscope reading by k. The second formula gives R where it matters, at the contact with the surface. Compare it with $R = C/2\pi$ from the circumference; they should agree within about 0.1 mm.

Uncertainty: the pen line lands on the mark within about 1 mm at each end, so $\delta\Phi \approx \sqrt{2} \cdot 1\,\text{mm}/R = 0.027\,\text{rad}$ out of $6\pi = 18.85\,\text{rad}$, which is 0.14 %. With $\delta d = 2\,\text{mm}$, $\delta R = 2\,\text{mm}/6\pi = 0.1\,\text{mm}$. Three rolls, then mean and SDOM.

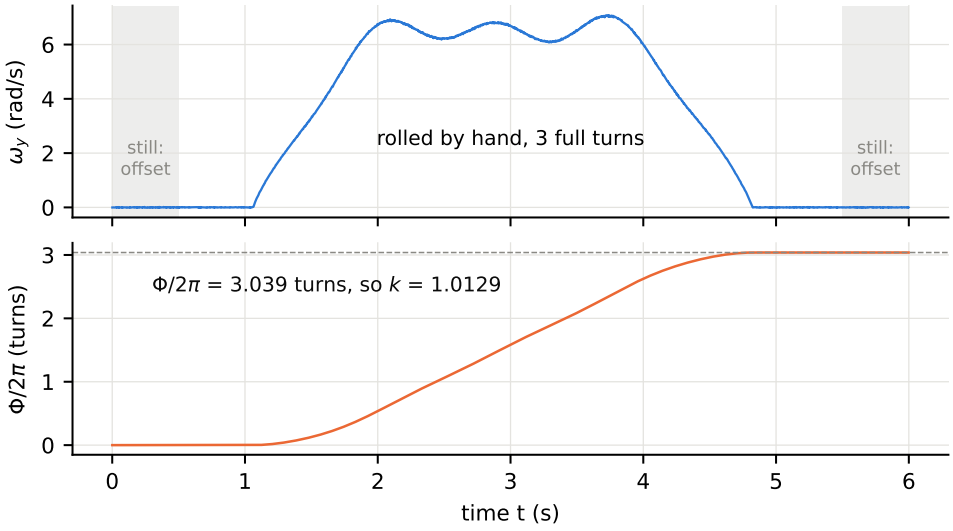


Figure 4. A calibration roll in the example data. Top: ω_y while the body is rolled by hand. Bottom: the angle $\Phi = \int \omega\, dt$ in turns. The gyroscope reports 3.039 turns for 3 real turns, so it reads 1.3 % too high.

5.8 The angle: accelerometer flips against an upright reference

In Experiment 3 your ruler gave 15.60° and the accelerometer 14.0° for the same ramp. On a 12° ramp, a 1.6° error is 13 % in $\sin\theta$, which times 18 wipes out the result. So the angle is measured against gravity, with one axis of the accelerometer only:

1. **Upright reference.** Hold the bare iOLab upright against a wall, y straight up, then upside down. The y axis reads $A_{y\uparrow} = +G + b_y$ and $A_{y\downarrow} = -G + b_y$. G is g as this axis sees it, scale error included; b_y is its offset.
2. **Flip on the board.** Lay the iOLab on its back on the board with y up the slope, then turn it around (same face down) so y points down the slope: $A_y = \pm G \sin\theta + b_y$.

$$\boxed{\sin\theta = \frac{A_{y,\mathrm{up}} - A_{y,\mathrm{down}}}{A_{y\uparrow} - A_{y\downarrow}}} \tag{14}$$

The offset cancels in each difference and the scale error cancels in the ratio. Because the reading is against gravity, a tilted table does not enter. Do the flip at three places along the board to see whether it bends.

Ruler check. $\sin\theta_{\mathrm{ruler}} = (h_{\mathrm{high}} - h_{\mathrm{low}})/L$ measures the angle against the table top. Do the same flip on the table (iOLab on its back, y toward the high end, then turned around) to get the tilt of the table itself, τ. Then $\theta = \theta_{\mathrm{ruler}} + \tau$. In the example data the ruler gives 12.18° and the accelerometer 12.41° ($z = 4.1$); the table tilts by $\tau = 0.23°$, and after the correction the two agree ($z = 0.1$).

Uncertainty. Each A_y is the mean of a still part: $\sigma/\sqrt{N}$ with $N = f\Delta t$. A tilt β of the upright iOLab lowers G by the factor $\cos\beta$; 2° gives 0.06 %. The spread of the three places gives the SDOM.

5.9 Uncertainty toolkit

These are the rules from Experiment 1, the course notebook on uncertainties and the Experiment 3 guide, plus three new ones (U7, U8, U11). Name them in the method section.

U1 *One sample.* A single reading has the uncertainty σ, the standard deviation of the signal on a still stretch.

U2 *Mean of a selection in the app.* Value μ, uncertainty $\sigma/\sqrt{N}$ with $N = f\Delta t$.

U3 *Repeated runs.* $\bar{x} = \frac{1}{n}\sum x_i$, $s = \sqrt{\sum(x_i - \bar{x})^2/(n-1)}$, $\delta\bar{x} = s/\sqrt{n}$ (SDOM). Use the SDOM for every mean over runs. If it comes out smaller than the single run uncertainty over $\sqrt{n}$, use that instead (the notebook does this).

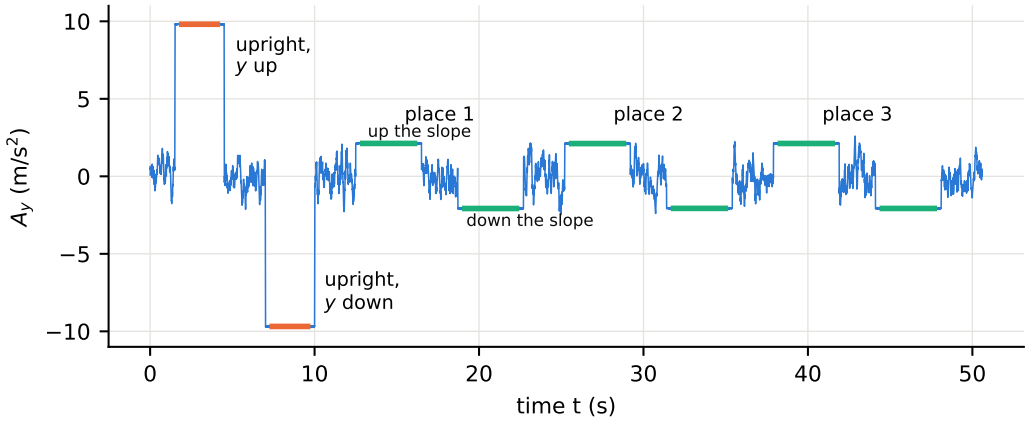


Figure 5. The angle recording in the example data, A_y against time. Orange: the upright reference. Green: the iOLab on its back on the board at three places, y up the slope and then down the slope. Everything else is handling; the notebook finds the still parts by their low noise.

U4 *Sum or difference.* $\delta(x \pm y) = \sqrt{\delta x^2 + \delta y^2}$.

U5 *Product or quotient.* $\delta z/|z| = \sqrt{(\delta x/x)^2 + (\delta y/y)^2}$. A power: $\delta(x^n)/|x^n| = |n|\,\delta x/|x|$.

U6 *Function of one variable.* $\delta y = |f'(x)|\,\delta x$. Example: $\theta = \arcsin(s)$ gives $\delta\theta = \delta s/\cos\theta$ in radians.

U7 *General rule, many inputs.* $\delta f = \sqrt{\sum_i \left(\frac{\partial f}{\partial x_i}\,\delta x_i\right)^2}$ for independent inputs. Each term is one line of the uncertainty budget. Section 5.10 writes out the derivatives.

U8 *Slope of a straight line fit.* $\delta b = s_{\text{res}}/\sqrt{\sum(t_i - \bar{t})^2}$, with s_{res} the standard deviation of the points around the line. The app shows the slope s but not its uncertainty; then use the SDOM over runs.

U9 *Time and value at a time.* A time read on a graph is uncertain by one sample, $\delta t = 1/f$. A value read at that time: $\delta y = \sqrt{\sigma^2 + (\dot{y}\,\delta t)^2}$.

U10 *Standard score.* $z = |x_1 - x_2|/\sqrt{\delta x_1^2 + \delta x_2^2}$: $z \le 1$ very likely the same, $1 < z \le 2$ likely the same, $2 < z \le 3$ grey zone, $z > 3$ unlikely the same.

U11 *Two results that share inputs.* M, R, $\sin\theta$, k and I_{rolls} go into both methods. An error in $\sin\theta$, M or I_{rolls} moves both results by the same amount and cancels in their difference. So compare the methods with the uncertainty of the difference, built only from the terms that differ (U7 applied to $I_N - I_E$). The notebook does it.

U12 *Scale and ruler readings.* A digital scale is uncertain by its last digit (0.1 g on a lab scale); a ruler by about half a millimetre for a reading, more if you also have to place it by eye.

U13 *Rounding.* Round the uncertainty to one significant figure (two if it starts with 1), the value to the same decimal place, and write the unit: $I_{\text{iOLab}} = (1.39 \pm 0.10) \times 10^{-4}\,\text{kg m}^2$.

5.10 The uncertainty of I_{iOLab}, written out

With the gyroscope readings divided by k, Method N is

$$I_{\text{iOLab}} = \frac{MgR\sin\theta\,k}{\bar{\alpha}_{\text{meas}}} - MR^2 - I_{\text{rolls}},$$

and Method E is

$$I_{\text{iOLab}} = \frac{2Mg\,(D_{\text{gate}} + R)\sin\theta\,(1 - \varepsilon - \Delta f)\,k^2}{\omega_{B,\text{meas}}^2} - MR^2 - I_{\text{rolls}}.$$

The partial derivatives for rule U7, written with I_P to keep them short:

Input	$\partial I_{\text{iOLab}}/\partial x$, Method N	$\partial I_{\text{iOLab}}/\partial x$, Method E
$\sin\theta$	$I_P/\sin\theta$	$I_P/\sin\theta$
$\bar{\alpha}$ or ω_B	$-I_P/\bar{\alpha}$	$-2I_P/\omega_B$
k	I_P/k	$2I_P/k$
R	$(I_P - 2MR^2)/R$	$I_P/s_{AB} - 2MR$
M	I/M	I/M
D_{gate}	(not used)	I_P/s_{AB}
ε	(not used)	$-I_P/(1-\varepsilon)$
I_{rolls}	-1	-1

Worked example, Method N with the example data. $I_P = 2.550 \times 10^{-3}\,\text{kg m}^2$. Each term is $|\partial I_{\text{iOLab}}/\partial x|\,\delta x$:

Input	value and uncertainty	term ($10^{-6}\,\text{kg m}^2$)
roll model (core lighter than tape)	shift $0.07 \times 10^{-4}\,\text{kg m}^2$	6.8
$\bar{\alpha}$ (7 runs, SDOM)	$(25.06 \pm 0.04)\,\text{rad/s}^2$	4.5
$\sin\theta$	0.2149 ± 0.0003	3.5
I_{rolls} (scale and radii)	$(7.99 \pm 0.03) \times 10^{-4}\,\text{kg m}^2$	3.4
k	1.0113 ± 0.0009	2.2
R	$(53.26 \pm 0.06)\,\text{mm}$	0.8
M	$(0.5682 \pm 0.0001)\,\text{kg}$	0.2
air drag left in $\bar{\alpha}$	up and down speeds differ a little	0.4
total, $\sqrt{\sum \text{term}^2}$		9.8

So $I_{\text{iOLab}} = (1.39 \pm 0.10) \times 10^{-4}\,\text{kg m}^2$. Method E gives $(1.43 \pm 0.12) \times 10^{-4}\,\text{kg m}^2$; there the largest terms are the roll model, D_{gate}, air drag and k.

6 The iOLab: settings that matter today

6.1 Start and calibration

1. Take the USB dongle out of the iOLab (press the metal end), plug it into the laptop, switch the iOLab on with the side button.
2. Press Reset twice. In the Tools menu, run the Acc/Magn/Gyro calibration and follow the screen. It asks for six still positions (each axis up, then down). Do it on a level table and hold each position still: this also sets the scale of each accelerometer axis. For the gyroscope it only sets the zero.
3. You do not need the force probe today.

6.2 The gyroscope

- Tick only Gyroscope. Fewer sensors give higher sample rates. Write the rate from the chart title in the log; expect 95 to 760 Hz.
- Unit: rad/s. If the numbers look 57 times too large, the file is in degrees per second. The notebook spots this from the calibration roll and converts.
- Range: the sensor can run at $\pm$250, $\pm$500 or $\pm$2000 degrees per second, which is $\pm$4.4, $\pm$8.7 or $\pm$34.9 rad/s. A curve that goes flat at the top has hit the limit; that run is useless. Keep ω below 90 % of the limit. With $R = 5.3$ cm, the largest range allows speeds up to 1.6 m/s, more than you need. If your test run shows a lower limit, use a smaller angle and shorter runs (Section 7.2).
- Data smoothing: off.

6.3 Axis and sign check, two minutes

Roll the body along the table toward you. ω_y is large; ω_x and ω_z stay near zero. If another axis is large, the rolls are not on the ends of the y axis: stop and check. Write down the sign of ω_y for rolling in one direction. The notebook turns the sign so that rolling down the ramp is positive.

6.4 Reading numbers in the app

Click and drag over a region with the graph tool active. The box shows μ (mean), σ (standard deviation), a (area under the curve), s (average slope, R^2 in brackets) and the length Δt. Zoom with the magnifier, double click to zoom out. Three uses today:

- Area of ω_y = angle turned, in rad. Remove the offset: $\Phi = a - \mu_{\text{still}}\,\Delta t$.
- Slope of ω_y over a window = α in rad/s^2.
- On a straight rise, the value at the end of a selection is $\mu + s\,\Delta t/2$ (the mean sits at the middle of the window).

6.5 Saving and exporting

- Every recording is saved by itself in `Documents/iOLab-WorkFiles/rawdata`. The folder icon opens old runs.
- To export: toolbar icon at the bottom left of the chart, then Export to CSV. The file goes to `Documents/iOLab-WorkFiles/export`. The columns are `time`, `raw[0..2]` and `cal[0..2]`; `cal` holds the calibrated x, y, z values.
- Right after each run, move the file into its own folder: `data/calibration/roll1`, `data/newton/run3`, `data/energy/run2`, and so on. The two accelerometer files go into `data/angle` and `data/table`. It takes 20 seconds and keeps runs apart.

7 Step by step

7.1 Measure the parts

Quantity	Tool	How	Uncertainty
m_{iOLab}	lab scale	iOLab with batteries, dongle out	last digit, 0.1 g
m_1, m_2	lab scale	each roll by itself	0.1 g
M	lab scale	the whole body with the rolls on; check against the sum	0.1 g
r_1, r_2	caliper (inside jaws) or ruler	inner diameter of the core in two directions; halve the mean	0.1 mm caliper, 0.5 mm ruler
R_1, R_2	paper strip and ruler	wrap the strip once around the roll, mark the overlap, lay it flat and measure C; $R = C/2\pi$	$\delta C = 1$ mm, so $\delta R = 0.16$ mm
w, t	ruler or caliper	width (x) and thickness (z) of the iOLab	0.5 mm
core wall	ruler	thickness of the cardboard ring (for the roll model)	0.5 mm
body length	ruler	outside edge to outside edge along the axis (air drag)	2 mm

Do not measure R with a ruler across the diameter: ± 1 mm there is 2 % in R, and R enters Method E strongly.

7.2 Build the ramp

- A straight, stiff board at least 0.8 m long and wider than the body, with a hard, clean surface. Lower end resting on the table, upper end on books. Mark a run line down the middle with tape.
- Angle 10° to 15°: a rise of 17 to 26 cm per metre of board. A steeper ramp makes the angle error smaller in relative terms; above about 20° the rolls can start to slip and the runs get short. If your gyroscope range turns out to be only $\pm 8.7\,\text{rad}/\text{s}$, use about 5° and runs of 15 cm; with $\pm 4.4\,\text{rad}/\text{s}$, 2° to 3° and runs of 10 cm, and expect a larger uncertainty.
- The end of the board at B must be a clean edge: the body leaves the board there. Put the towel on the table 20 to 40 cm beyond the end.
- Once the angle is measured, nobody touches the ramp. If it moves, measure the angle again.

7.3 Measure the angle (one recording, about a minute)

Tick only Accelerometer and press Record.

1. Hold the bare iOLab upright against a wall or the side of a box, y straight up. Use the set square to keep it vertical. Hands still, 4 s. Then upside down, y straight down, 4 s.

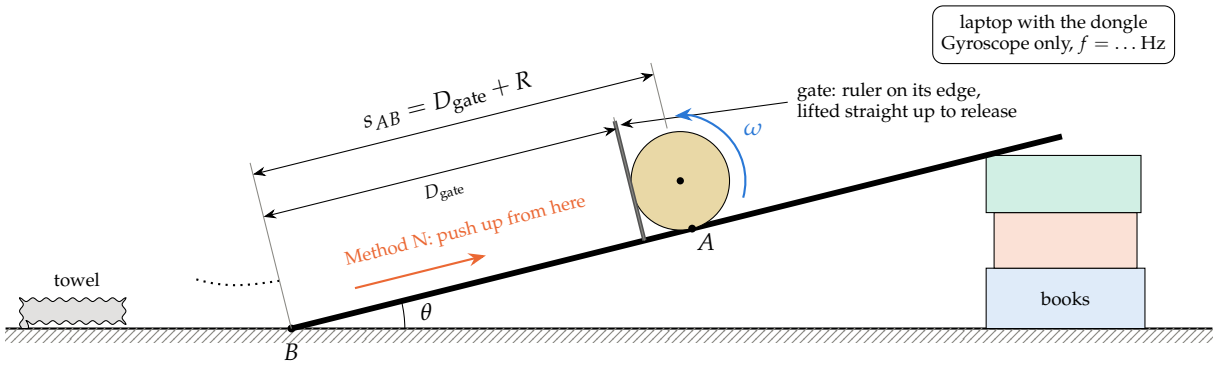


Figure 6. Setup seen from the side, angle drawn larger than real. Method E: the body rests against the gate with its contact line at A, rolls the distance s_{AB} to the end of the board B, leaves the board and lands on the table. Method N: the body is pushed up the board from near the bottom and caught on its way back, before B.

2. Lay the iOLab on its back on the board at place 1 (near the bottom), its long side parallel to the board edge, y pointing up the slope. Hands off, 4 s.
3. Turn it around, same face down, y pointing down the slope. 4 s.
4. Repeat at place 2 (middle) and place 3 (near the top). Press Stop. Export the accelerometer chart into `data/angle`.

Table check, second recording: lay the iOLab on its back on the table next to the ramp, y pointing toward the high end of the ramp, 4 s; turn it around, 4 s. Export into `data/table`.

Ruler check: put two marks on the board a distance L apart (for example 0.800 m). Measure the height of the top surface of the board above the table at both marks, ruler held vertical against the set square. Write L, h_{low}, h_{high}.

7.4 Mount the rolls

- Compare the circumferences first. Pick two rolls within 2 mm of each other (0.3 mm in R). Different radii make the body roll in a curve.
- Press each roll onto an end of the iOLab up to the small notch, not past it. Same depth at both ends.
- Press the loose end of the tape flat on each roll so it cannot catch or make a bump once per turn.
- Weigh the whole body: M. It should equal the sum of the three parts within the scale step.
- Roll the body slowly along the table. It should run straight, the iOLab should not wobble, and ω_y should be smooth. A ripple once per turn means the iOLab sits off centre in a core: turn that roll a little on the iOLab end and try again.

7.5 Calibration rolls, three times

1. Draw a thin pen line across the outer surface of roll 1. Stick masking tape on the table and draw a start line on it.
2. Set the body down with the pen line touching the start line. Tick Gyroscope only. Record; 1 s still.
3. Push the body slowly along the table with one finger on the middle of the iOLab, straight, for exactly 3 turns, until the pen line touches the table for the third time. Stop there and mark the table. 1 s still, then Stop.
4. Measure d between the two marks with the tape measure. Export to `data/calibration/roll1`. Write d in the log.

Roll at about one turn per second so nothing slips. Repeat three times. Expect $d \approx 1.0$ m for $R = 5.3$ cm.

7.6 Method N runs, up and down

1. Body at rest on the run line about 10 cm above the end, axis square to the board edge. Record; 1 s still.
2. Push it up the board with one smooth push of a finger on the middle of the iOLab, then take the hand away at once. Aim for a trip of 30 to 50 cm up the board, which means ω about 15 to 22 rad/s after the push.

3. Let it roll up, stop and roll back. Catch it before it reaches the end. Stop.
4. Look at the run: two straight lines that cross zero once (Figure 7). Export to `data/newton/run1` and so on. Log it.

Get 6 to 8 good runs. Redo a run if the body drifts sideways more than about 2 cm, hits anything, or the graph is not two clean straight lines.

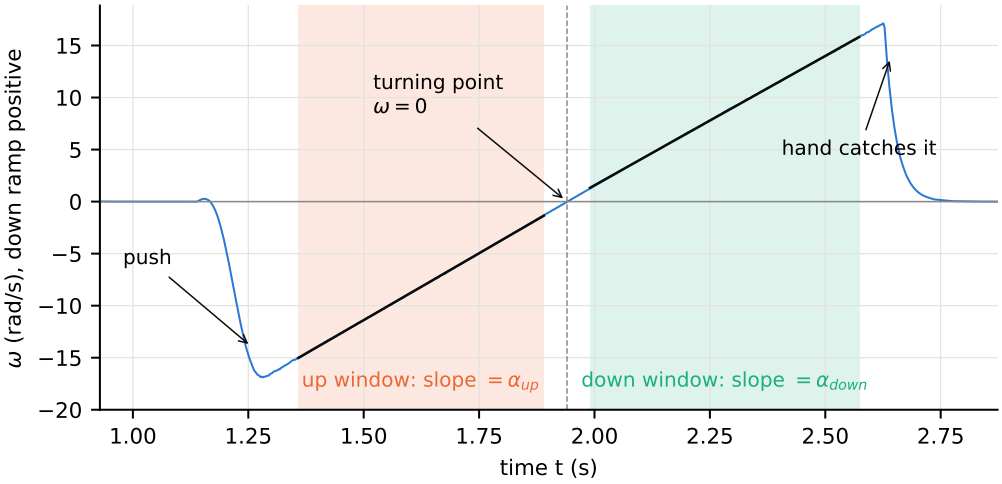


Figure 7. A Method N run in the example data. After the push, ω rises as a straight line through zero at the turning point. The fitted slopes in the two windows are α_{up} and α_{down}; the windows skip the push, the turning point and the catch.

7.7 Method E runs, release at the gate

1. Put a tape mark about 45 cm up the board from the end. Hold a ruler on its edge across the board at the mark, square to the board surface. Rest the body against it from above, axis square to the board edge.
2. Once: measure D_{gate} along the board surface from the gate face to the end B. Uncertainty about 1 mm. Then $s_{AB} = D_{\text{gate}} + R$.
3. Record; 1 s still with the body resting against the gate.
4. Lift the ruler straight up, fast. Do not push the body.
5. The body rolls down, leaves the board at B, lands on the table and runs into the towel. Stop.
6. Look at the run: a straight rise from zero, a sharp corner, a small drop (Figure 8). Export to `data/energy/run1` and so on.

Get 6 to 8 runs from the same gate mark. Why 45 cm: a longer run means more air drag and a faster body; a shorter run makes the 1 mm of D_{gate} count more. 40 to 60 cm works. Below about 30 cm the body is too slow to fly off the edge at once (v_B^2/R must be larger than $g\cos\theta$, so $v_B > 0.75\,\text{m/s}$); it then tips over the edge and the corner gets blurred.

7.8 What to read and how, if you work in the app

The notebook finds all of these by itself. As a backup, or to check it, read them like this and type them into the manual mode of the notebook:

Quantity	Graph	How
offset b	ω_y	still first second: μ
Φ (calibration)	ω_y	select from the still part before to the still part after: $\Phi = a - b\,\Delta t$
α_{up}	ω_y	from about 0.1 s after the push to 0.05 s before $\omega = 0$: the slope s
α_{down}	ω_y	from 0.05 s after $\omega = 0$ to 0.05 s before the catch: s
ω_B	ω_y	select the last 0.15 s of the rise, ending exactly at the corner: $\omega_B = \mu + s\,\Delta t/2$. Or select the flat flight part: μ
A_y for the angle	A_y	each still part: μ, σ, Δt

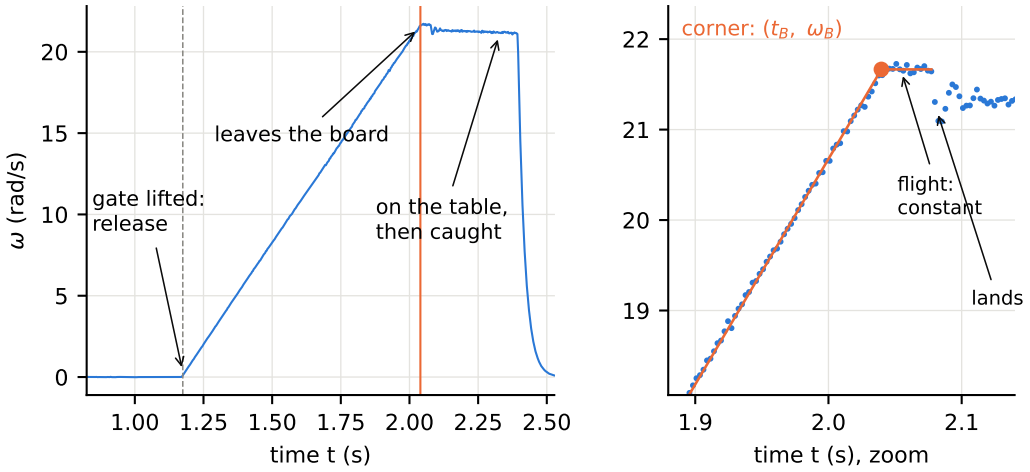


Figure 8. A Method E run in the example data. Right: zoom on the end of the board. The orange hinge (a straight line, then flat) is fitted to the data; its corner is (t_B, ω_B). After the corner the body flies for about 0.04 s at constant ω, then lands on the table.

7.9 Rough values to catch mistakes

These are ranges for spotting errors, not targets.

Quantity	Rough value
gyroscope at rest	offset below 0.05 rad/s, noise about 0.01 rad/s
k	0.97 to 1.03
R	4.5 to 6 cm; from the roll and from the circumference within 0.2 mm
M	0.4 to 0.8 kg
$\sin\theta$	0.17 to 0.26 for 10° to 15°; the three places within 0.001
$\bar{\alpha}$	about $g \sin\theta/(1.6R)$: 20 to 35 rad/s^2
ε	0.005 to 0.03
ω_B	15 to 30 rad/s
$I/(MR^2)$	0.5 to 0.7
I_j for one roll	3 to 5 $\times 10^{-4}$ kg m^2
I_{iOLab}	1.0 to 1.9 $\times 10^{-4}$ kg m^2

If I_{iOLab} comes out negative or above 3×10^{-4} kg m^2, look at the angle first, then at k, then at R.

7.10 Mistakes that cost points

- Measuring the angle with a ruler only. Any table tilt goes straight into $\sin\theta$ (Experiment 3).
- No calibration roll. A 1 % scale error of the gyroscope is 18 to 36 % in I_{iOLab}.
- Method N from release runs only. Friction then stays in α.
- Method E without ε: 31 % too large.
- R from a ruler across the diameter, or s_{AB} without the $+R$.
- Windows that include the push, the turning point or the catch.
- A body that drifts sideways or touches the edge of the board.
- Final numbers without the run tables, or I_{iOLab} without its uncertainty.
- Units: I in kg m^2, α in rad/s^2, ω in rad/s.

8 The analysis notebook

`Experiment4_analysis.ipynb` does every calculation in this guide on your exported files. It was tested on simulated iOLab files in three export layouts and in manual mode. It prints numbers, tables and figures. It

writes no report text.

Before the lab, 10 minutes. Keep the notebook and the `example_data` folder together in your `4th_experiment` folder. Run all cells once. It analyses the simulated example data and prints every step; the dummy report uses these numbers. If a package is missing, run `pip install numpy pandas matplotlib` in a terminal. Then make the folders `data/calibration`, `data/angle`, `data/table`, `data/newton` and `data/energy`.

In the lab.

1. One folder per run (`roll1`, `run1`, ...) with its CSV file inside. File names do not matter.
2. In the settings cell, set `DATA_DIR = Path("data")` and type your numbers: masses, `D_in`, `C_out`, `CAL_DIST` (one per roll), `RULER`, `D_GATE`, `BODY_LENGTH`, `CORE_WALL`.
3. Run all cells. Look at the plot of every run: the windows in Method N, the corner in Method E.
4. If a window or a corner is wrong, set it by hand and run again, for example `MANUAL_WINDOWS = {"run3": {"up": (1.40, 1.85), "down": (2.00, 2.55)}}` or `MANUAL_CORNER = {"run2": 2.153}`.

What it gives you. Step 1, the parts and I_{rolls}. Step 2, k and R. Step 3, $\sin\theta$ with the ruler and table checks. Step 4, Method N with a table of runs, ε, I_P, I, I_{iOLab} and the budget. Step 5, Method E with the same, plus the two free checks. Step 6, the comparison, the combined value, the box model range and two report figures in `output`: `report_figure_runs` and `report_figure_results`.

If the CSV export fails. Read the numbers in the app (Section 6.4 and the table in Section 7.8) and type them into `MANUAL_CAL`, `MANUAL_ANGLE`, `MANUAL_NEWTON` and `MANUAL_ENERGY`. The settings cell shows the format. The analysis is then the same.

If a file reads wrongly. The notebook prints how it read each file. Set `GYRO_AXIS = "y"` if it picks the wrong axis, and `GYRO_UNITS` to `"rad/s"` or `"deg/s"` if the unit check fails.

9 Report blueprint

The rubric rewards clear evidence, not length, and three pages fill up fast. This section says what goes where and what each part must contain. The dummy report shows one way to do all of it.

9.1 Page plan

Page	Content
1	Title block. Purpose (3 lines). Setup: the sketch next to 5 to 6 lines of text. Theory: (3), (6), (7), (8), the parallel axis theorem and (11).
2	Method: calibration roll, angle, both run types, how the numbers come out of the data, the uncertainty rules by name. Table 1 with the measured inputs. The run tables.
3	Figure with one run of each method. Results with uncertainties. Comparison between the methods and with the box model. Discussion: the lever (5 %, times 18), the largest budget terms, error sources with their direction. Conclusion.
4	Reflections only.

Space savers: sketch and text side by side; one figure with two panels; small font inside tables; do not copy the task text.

9.2 What each section must contain

Title block
☐ A title that says what you measured, not only "Experiment 4".
☐ Names, study numbers, group number, date, course number.

Purpose
☐ One sentence on the problem: the iOLab adds to the moment of inertia of the model wings, so its own I

about y is needed to correct the measurements.
☐ One sentence on what you did: two independent determinations, energy and torque.

Setup
☐ The sketch (Section 9.3).
☐ The body: iOLab plus two tape rolls, the rolling axis is y, the gyroscope measures ω_y, its rate.
☐ The ramp: board, θ, the gate and D_{gate}, the end B, where the body lands.
☐ Masses and radii with uncertainties (or in Table 1).

Theory
☐ Rolling without slipping and $K = \frac{1}{2} I_P \omega^2$ with the parallel axis theorem.
☐ Method N: torque about P, the up and down equations, why friction cancels.
☐ Method E: the energy balance with the friction loss.
☐ $I_{\text{iOLab}} = I_P - MR^2 - I_{\text{rolls}}$ and the hollow cylinder formula.

Method
☐ Each run type in 2 to 3 sentences: start, push or release, stop.
☐ Numbers of runs, and why you repeated.
☐ The calibration roll and the angle method in one or two sentences each.
☐ How the quantities came out of the data: straight line fits in windows, the corner, CSV files analysed in Python.
☐ The uncertainty rules by name: $\sigma/\sqrt{N}$, SDOM, propagation with partial derivatives, standard score.

Results
☐ Table 1 with every measured input. Run tables with every run and the mean.
☐ One figure with axis labels, units, windows and corner, and a caption that says which run and what to look at.
☐ I_P, I and I_{iOLab} for both methods, with uncertainty and unit, in the text.

Discussion and conclusion
☐ The two methods against each other: z from the uncertainty of the difference (U11).
☐ The result against the box model range from (12).
☐ Why the uncertainty is what it is: the iOLab is 5 % of I_P, and the largest budget terms.
☐ Two or three error sources with the direction each one pushes the result (Section 9.6).
☐ The final statement: $I_{\text{iOLab}} = (\ldots \pm \ldots) \times 10^{-4}\,\text{kg m}^2$, and what it means for the windmill model.

9.3 The sketch

Draw it by hand. Figures 1 and 6 show the level of labelling.

- Side view: ☐ table ☐ board and books ☐ θ ☐ gate and D_{gate} ☐ contact line A and end B ☐ s_{AB} ☐ towel ☐ the body as a circle with its axis ☐ ω and the rolling direction ☐ the push direction for Method N ☐ the laptop with "Gyroscope, $f = \ldots$ Hz".
- Front view: ☐ iOLab between the two rolls ☐ notches ☐ r and R ☐ the y axis along the rolling axis ☐ the contact line.

Photo: daylight, camera straight above the paper, crop the edges, darken the lines if needed. The text in the sketch must be readable at the size you insert it.

9.4 Table templates

Table 1, measured inputs: one row per quantity with value, uncertainty, unit and how you got it (m_{iOLab}, m_1, m_2, M, r_1, r_2, R_1, R_2, R from the roll, k, $\sin\theta$, θ, D_{gate}, s_{AB}, I_{rolls}).

	Method N					Method E		
Run	α_{up} (rad/s^2)	α_{down} (rad/s^2)	ε	I_{iOLab} (10^{-4} kg m^2)	Run	ω_B (rad/s)	I_P (10^{-3} kg m^2)	I_{iOLab} (10^{-4} kg m^2)
1					1			
⋮					⋮			
mean					mean			

Units in the header, not in the cells. The per run I_{iOLab} shows the scatter; the final value comes from the means and the full budget.

9.5 Figures

☐ Axis labels with symbol and unit, for example "ω (rad/s)" and "time t (s)".
☐ Windows shaded, fitted lines and the corner marked.
☐ A caption of about two lines: what is shown, which run, what the reader should see.
☐ The text refers to every figure.

9.6 Error sources worth one sentence each

Source	Effect and direction
Angle	$\sin\theta$ too large gives I_P and I_{iOLab} too large, times 18. Controlled by the flips against the upright reference; ruler and table check agree.
Gyroscope scale	Measured with the calibration roll; what is left is about 0.1 %.
Rolling resistance	Cancels in Method N. Taken out by ε in Method E; without it, Method E is 31 % too large.
Air drag	Makes Method E slightly too large. Corrected by Δf, half of it kept as uncertainty.
Tape roll model	The cardboard core is lighter than tape, so (11) gives I_{rolls} about 1 % too small and I_{iOLab} about $0.07 \times 10^{-4}\,\text{kg m}^2$ too large.
Slipping	Would make ω too small for the distance. The check $R\Phi = s_{AB}$ shows no slip.
iOLab off centre	A ripple once per turn; it averages out over two or more turns. The extra me^2 is below $10^{-6}\,\text{kg m}^2$ for $e < 2\,\text{mm}$.
Axis not square	The driving torque shrinks by $\cos\psi$; 3° gives 0.14 %, so redo runs that drift.
Board bends	The three places of the angle recording show it.

9.7 Page 4: reflections

Three to four lines per question. Be concrete and name a moment from the day. These are prompts, not answers to copy.

1. *How did you approach the experiment: expectations, or only the data?* Did you estimate I_{iOLab} from the box model before the lab? Did you expect the iOLab to be such a small part of I_P? Did you predict that $\alpha_{\text{up}} > \alpha_{\text{down}}$?
2. *What did you learn?* One or two concrete things. For example: calibrating a sensor against a motion you know (three turns); cancelling friction by going up and down; how a small difference of large numbers blows up the uncertainty; that the angle and the gyroscope scale matter more than the number of runs.
3. *What were you supposed to learn?* Your view of the aim: rotational dynamics of a rolling rigid body with energy and with torque, the parallel axis theorem, two independent determinations of one quantity, and propagation of uncertainty through a subtraction.

9.8 Final check before you submit

☐ At most 3 pages, then page 4.
☐ Names, study numbers, group, date, course.
☐ Purpose with the windmill reason in one sentence.
☐ Hand drawn sketch with both views, labelled and readable.
☐ Every number with uncertainty and unit, rounded by rule U13.
☐ Table 1 with the inputs; run tables with means and SDOM.
☐ Figure with axis labels, units and a caption, referred to in the text.
☐ Two values of I_{iOLab}, a z between them, a comparison with the box model.
☐ Sample rate, number of runs, k and the angle method stated.
☐ Reflections: three questions, 3 to 4 lines each.
☐ The same symbols everywhere (I_P, I, I_{rolls}, I_{iOLab}, α_{up}, α_{down}, ω_B, ε).
☐ Exported to PDF; figures sharp.

A Formula sheet

Rolling without slipping	$v = \omega R$, $a = \alpha R$, $K = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}I_P\omega^2$, $I_P = I + MR^2$
Method N	$I_P\alpha_{\text{up}} = MgR\sin\theta + F_rR$, $I_P\alpha_{\text{down}} = MgR\sin\theta - F_rR$
	$I_P = MgR\sin\theta/\bar{\alpha}$, $\bar{\alpha} = \frac{1}{2}(\alpha_{\text{up}} + \alpha_{\text{down}})$, $\varepsilon = (\alpha_{\text{up}} - \alpha_{\text{down}})/(\alpha_{\text{up}} + \alpha_{\text{down}})$
Method E	$I_P = 2Mg\,s_{AB}\sin\theta\,(1 - \varepsilon - \Delta f)/\omega_B^2$, $s_{AB} = D_{\text{gate}} + R$
Air drag	$k_d = \frac{1}{2}\rho C_D A$, $A = 2R\cdot$ body length; $\Delta f = k_d v_B^2/(2Mg\sin\theta) - k_d\langle v^2\rangle_N/(Mg\sin\theta)$
Body and iOLab	$I = I_P - MR^2$, $I_{\text{iOLab}} = I - I_{\text{rolls}}$
Tape rolls	$I_j = \frac{1}{2}m_j(r_j^2 + R_j^2)$, $R_j = C_j/2\pi$
Box model	$\frac{1}{12}m(w^2 + t^2) \le I_{\text{iOLab}} \le \frac{1}{12}m(w + t)^2$
Calibration roll	$k = \Phi/2\pi N$, $R = d/2\pi N$; divide every ω and α by k
Angle	$\sin\theta = (A_{y,\text{up}} - A_{y,\text{down}})/(A_{y\uparrow} - A_{y\downarrow})$; ruler: $\theta = \theta_{\text{ruler}} + \tau$
App readings	$\Phi = a - b\,\Delta t$; value at the end of a straight selection $\mu + s\Delta t/2$
Uncertainty	$\sigma/\sqrt{N}$ with $N = f\Delta t$; SDOM $= s/\sqrt{n}$; $\delta f = \sqrt{\sum(\partial f/\partial x_i\,\delta x_i)^2}$
	slope: $\delta b = s_{\text{res}}/\sqrt{\sum(t_i - \bar{t})^2}$; time: $\delta t = 1/f$
Partial derivatives	N: $I_P/\sin\theta$, $-I_P/\bar{\alpha}$, I_P/k, $(I_P - 2MR^2)/R$, I/M, -1
	E: $I_P/\sin\theta$, $-2I_P/\omega_B$, $2I_P/k$, $I_P/s_{AB} - 2MR$, I/M, I_P/s_{AB}, $-I_P/(1-\varepsilon)$, -1
Standard score	$z = \lvert x_1 - x_2\rvert/\sqrt{\delta x_1^2 + \delta x_2^2}$: ≤ 1 very likely the same, ≤ 2 likely, ≤ 3 grey zone, > 3 unlikely
Constants	$g = 9.82\,\text{m/s}^2$ (the course value), $\rho_{\text{air}} = 1.2\,\text{kg/m}^3$

B Run log and measurement sheet

Copy this onto paper. One line per recording, bad runs included. It saves you in the analysis.

Type	Run	File time stamp	Good?	Notes (d, push, drift, what went wrong)
calibration	1			$d =$
calibration	2			$d =$
calibration	3			$d =$
angle				places 1, 2, 3 upright: yes / no
table				
Method N	1			
Method N	2			
Method N	3			
Method N	4			
Method N	5			
Method N	6			
Method N	7			
Method N	8			
Method E	1			
Method E	2			
Method E	3			
Method E	4			
Method E	5			
Method E	6			
Method E	7			
Method E	8			

$m_{\text{iOLab}} =$		$m_1 =$		$m_2 =$		$M =$	
$D_{\text{in},1} =$		$D_{\text{in},2} =$		$C_1 =$		$C_2 =$	
$w =$		$t =$		core wall $=$		body length $=$	
$L =$		$h_{\text{low}} =$		$h_{\text{high}} =$		$D_{\text{gate}} =$	
gyroscope rate $=$	Hz	accelerometer rate $=$	Hz	range limit $=$	rad/s	sign of ω_y toward you:	